Bayesian Inference of Soccer Expected Goals (xG)

Shin Adachi

April 16, 2026

Introduction

Expected Goals (xG) is a fundamental metric in modern soccer (football) analytics used to quantify the quality of a scoring opportunity. It estimates the probability of a goal for each shot based on the historical data. The higher that xG is, the more likely a team is to have scored that particular chance. While frequentist models often provide a single point estimate, this project utilizes a Bayesian framework to quantify the inherent uncertainty in match events. By treating model parameters as random variables, we can derive posterior predictive distributions for specific iconic scenarios, such as long-range free kicks or overhead strikes.

Data

The dataset consists of every shot recorded in 522 matches in La Liga across five seasons (2014/15 to 2018/19), available via StatsBomb [1] and imported using their native Python package *statsbombpy*.

The imported dataset contains 12,034 total shots with 1,349 successful goals (conversion rate $\approx 11.2\%$). It includes the shot location on the field, which was then used to calculate the Euclidean distance to the goal center and the angle (the visible width of the goal mouth) using the law of cosines. Despite variations in pitch dimensions due to loose regulation, the field size is assumed to be 120 by 80 yards for simplicity. The calculated distance and angle are used as the main features. The data also contains a *technique* feature describing the method used for each shot, such as volley, overhead kick, and backheel. Based on this information, a new binary variable was engineered to distinguish between "normal" shots and "special techniques."

The distance and angle features were standardized to a mean of zero and unit variance. The densities of both distance and angle are unimodal and approximately normally distributed (Figures 1 and 2). For distance, the majority of goals come from shorter-range shots, whereas the peak density of all shots is centered around zero. Counterintuitively, the conversion rates for shots with and without special techniques are approximately equal (Figure 3).

Model

Bayesian logistic regression was used to build two models: one using only distance and angle, and another with technique as an additional feature. The goal outcome for shot i, denoted y_i, is modeled as a Bernoulli

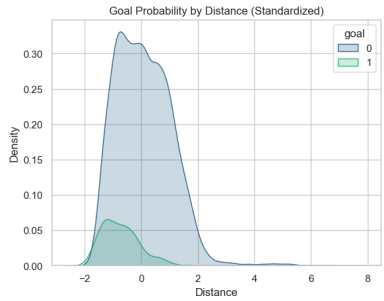

Figure 1: Goal Probability by Distance (Standardized)

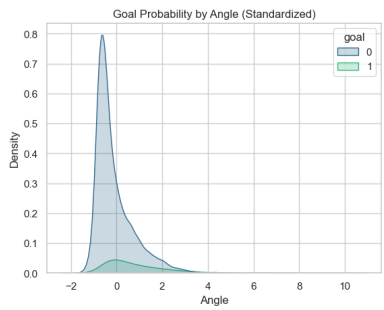

Figure 2: Goal Probability by Angle (Standardized)

random variable:

$$y_i \sim \text{Bernoulli}(p_i) \tag{1}$$

For the first model (spatial only), the probability p_i is mapped through the logit function:

$$\text{logit}(p_i) = \beta_0 + \beta_{\text{dist}} \cdot x_{i,\text{dist}} + \beta_{\text{angle}} \cdot x_{i,\text{angle}} \tag{2}$$

Similarly, for the second model:

$$\text{logit}(p_i) = \beta_0 + \beta_{\text{dist}} \cdot x_{i,\text{dist}} + \beta_{\text{angle}} \cdot x_{i,\text{angle}} + \beta_{\text{tech}} \cdot x_{i,\text{tech}} \tag{3}$$

where $x_{i,\text{dist}}$, $x_{i,\text{angle}}$, and $x_{i,\text{tech}}$ denote the standardized shot distance, standardized shot angle, and whether the shot was taken with a special technique for shot i, respectively.

Prior Distributions

To provide weakly informative regularization and avoid overfitting, we assign Normal priors to all coefficients:

$$\beta_j \sim \text{Normal}(\mu = 0, \sigma = 2)$$

This choice reflects the assumption that no single feature has an unbounded effect on goal probability, while allowing the large dataset ($n > 12{,}000$) to dominate the posterior.

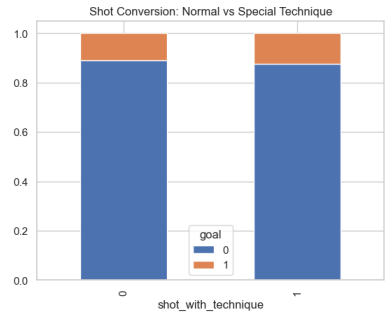

Figure 3: Shot Conversion: Normal vs Special Technique

Computational Details

The model was implemented in **Python** using the **PyMC** library [2]. We used the No-U-Turn Sampler (NUTS) [3] with 2,000 draws across 4 chains after 1,000 tuning steps.

Convergence Diagnostics

Convergence was verified using the $\hat{R}$ statistic and trace plots. All parameters in both models achieved $\hat{R} = 1.00$, indicating that the chains successfully reached the stationary distribution (Tables 1 and 2). The trace plots for both models show no evidence of poor convergence (Figure 6 and 7). The Effective Sample Size (ESS) was sufficiently high for all parameters (ESS $> 3{,}000$).

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
intercept	-2.455	0.040	-2.533	-2.376	0.001	0.000	3865.000	4205.000	1.000
beta_dist	-0.834	0.065	-0.967	-0.710	0.001	0.001	3052.000	3322.000	1.000
beta_angle	0.289	0.040	0.214	0.370	0.001	0.000	3530.000	4024.000	1.000

Table 1: Model 1 Posterior Summary

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
intercept	-2.471	0.044	-2.560	-2.387	0.001	0.001	4248.000	4883.000	1.000
beta_dist	-0.830	0.065	-0.955	-0.699	0.001	0.001	3200.000	3985.000	1.000
beta_angle	0.291	0.040	0.215	0.369	0.001	0.000	3634.000	4296.000	1.000
beta_technique	0.058	0.069	-0.073	0.193	0.001	0.001	5379.000	4740.000	1.000

Table 2: Model 2 Posterior Summary

Results

For Model1 (spatial only), the posterior mean estimate for β_{dist} is -0.834 with a 95% HDI of $[-0.967, -0.710]$, and for β_{angle} it is 0.289 with a 95% HDI of $[0.214, 0.370]$. These results confirm the widely held belief that shots from closer range and at a wider angle are more likely to be converted.

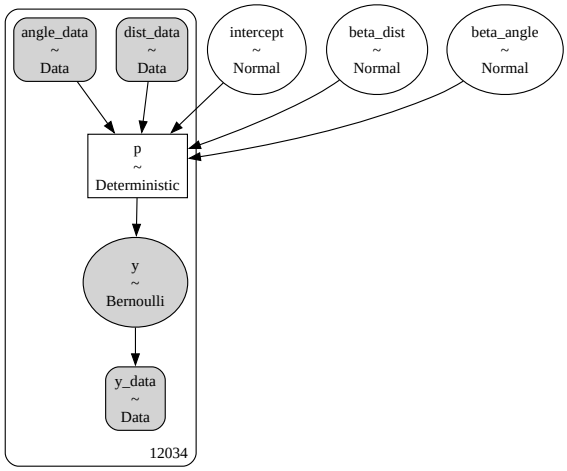

Figure 4: Model 1 Architecture

For Model2 (spatial + technique), the posterior mean for β_{dist} is -0.830 with a 95% HDI of $[-0.955, -0.699]$, for β_{angle} it is 0.291 with a 95% HDI of $[0.215, 0.369]$, and for β_{tech} it is 0.058 with a 95% HDI of $[-0.073, 0.193]$. This suggests that while the mean effect is slightly positive, we cannot conclude with 95% certainty that special techniques statistically increase scoring probability when controlling for distance and angle.

Scenario Analysis (Posterior Predictive)

To evaluate the practical utility of the Bayesian models, the following five distinct match scenarios were selected to represent a spectrum of scoring difficulty:

1. Back post tap-in (coordinates: (114, 30), technique: normal)
2. A shot from the penalty spot (coordinates: (108, 40), technique: normal)
3. Gareth Bale's overhead kick from around the penalty spot in the Champions League final in 2018 [4](coordinates: (108, 40), technique: overhead kick)
4. An edge-of-the-box shot from the left-winger [5](coordinates: (98, 24), technique: normal)
5. David Beckham's long-range free kick against Greece [6] (coordinates: (87, 40), technique: normal)

The back-post tap-in ($p[0]$) yielded a mean xG of 0.133 in Model 1 and 2, which is statistically lower than the intuitively expected conversion rate for such a clear scoring opportunity (Tables 3 and 4). This discrepancy

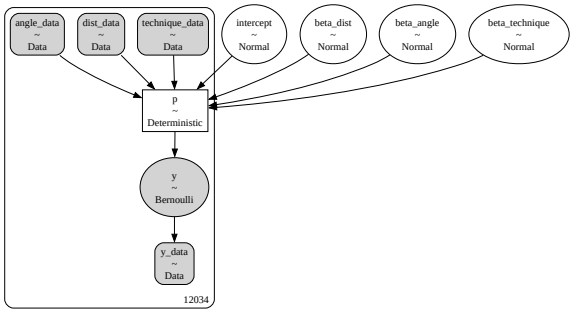

Figure 5: Model 2 Architecture

likely arises from the model's inability to account for the absence of a goalkeeper or defender pressure, features not present in the current dataset.

Comparing a standard shot from the penalty spot ($p[1]$) and Gareth Bale's overhead kick ($p[2]$) revealed a counterintuitive result in Model 2. While Model 1 assigned both shots an identical spatial xG of 0.166, Model 2 estimated a higher mean xG for the overhead kick (0.172) than the normal shot (0.163). As shown in Figures 8 and 9, the inclusion of the special technique feature in Model2 shifted the predictive density for this scenario rightward with greater variance. This is likely due to a selection bias in football, where such difficult maneuvers are typically attempted only when a player perceives a specific tactical advantage or a high-skill window of opportunity.

The model demonstrated high precision in identifying the statistical rarity of long-range efforts. The edge-of-the-box shot ($p[3]$) and David Beckham's free kick ($p[4]$) resulted in mean xG values of 0.028 and 0.016, respectively. The narrow 95% HDI for these scenarios, ranging from 0.012 to 0.020 for the Beckham free kick implies that such a goal is statistically improbable with high certainty.

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
p[0]	0.133	0.007	0.119	0.145	0.000	0.000	4688.000	5211.000	1.000
p[1]	0.163	0.005	0.154	0.174	0.000	0.000	6154.000	6243.000	1.000
p[2]	0.172	0.008	0.155	0.187	0.000	0.000	6618.000	6347.000	1.000
p[3]	0.028	0.002	0.023	0.032	0.000	0.000	3869.000	4684.000	1.000
p[4]	0.016	0.002	0.012	0.020	0.000	0.000	3557.000	4255.000	1.000

Table 3: Model 1: Posterior Predictive Distribution of Goal Probability for New Scenarios

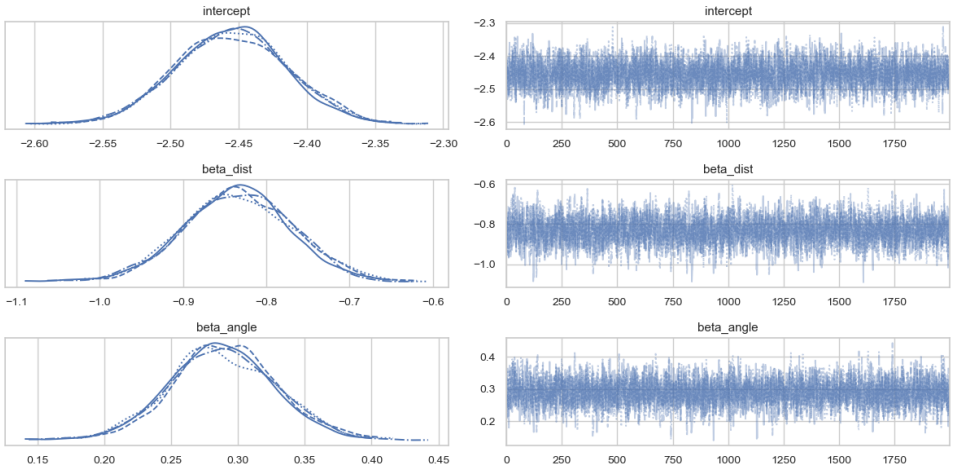

Figure 6: Model 1 Trace Plot

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
p[0]	0.133	0.007	0.119	0.145	0.000	0.000	4688.000	5211.000	1.000
p[1]	0.163	0.005	0.154	0.174	0.000	0.000	6154.000	6243.000	1.000
p[2]	0.172	0.008	0.155	0.187	0.000	0.000	6618.000	6347.000	1.000
p[3]	0.028	0.002	0.023	0.032	0.000	0.000	3869.000	4684.000	1.000
p[4]	0.016	0.002	0.012	0.020	0.000	0.000	3557.000	4255.000	1.000

Table 4: Model 2: Posterior Predictive Distribution of Goal Probability for New Scenarios

Conclusion

This project applied Bayesian logistic regression to over 12,000 shots from five seasons of La Liga data to model Expected Goals (xG) under a probabilistic framework. Two models were evaluated: a spatial model using shot distance and angle, and an extended model incorporating shot technique as an additional binary feature.

The posterior estimates confirmed that shot distance is the dominant predictor of goal probability, with β_{dist} is -0.834 (95% HDI $[-0.967, -0.710]$), while shot angle contributes positively, with β_{angle} (95% HDI $[0.214, 0.370]$). The technique coefficient in Model 2 (β_{tech}, 95% HDI $[-0.955, -0.699]$) spans zero, indicating that special techniques do not provide a statistically conclusive advantage in scoring probability when controlling for spatial factors. Both models achieved full convergence with $\hat{R} = 1.00$ and ESS $> 3{,}000$. all parameters.

The posterior predictive scenario analysis further illustrated the practical utility of the Bayesian approach. The model correctly assigned low xG values to long-range efforts such as David Beckham's free kick (xG=0.016) and the edge-of-the-box shot (xG=0.028), while the back-post tap-in and penalty spot scenarios received higher estimates. The counterintuitive underestimation of the tap-in highlights the limitation of such a simple model.

Overall, the Bayesian framework proved well-suited for sports analytics, providing not just point estimates but full posterior distributions that honestly quantify the uncertainty inherent in predicting rare, high-skill athletic outcomes.

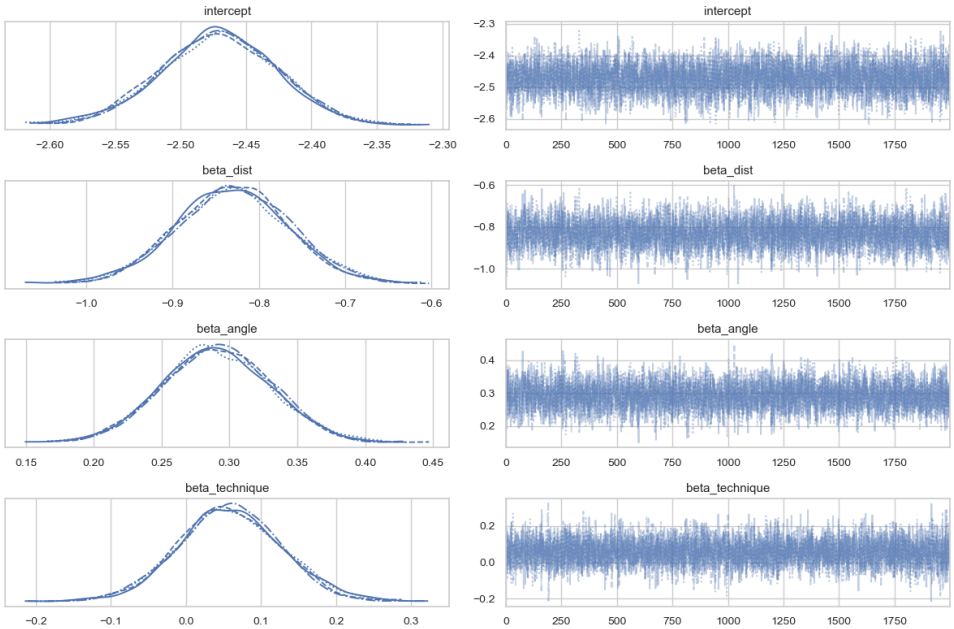

Figure 7: Model 2 Trace Plot

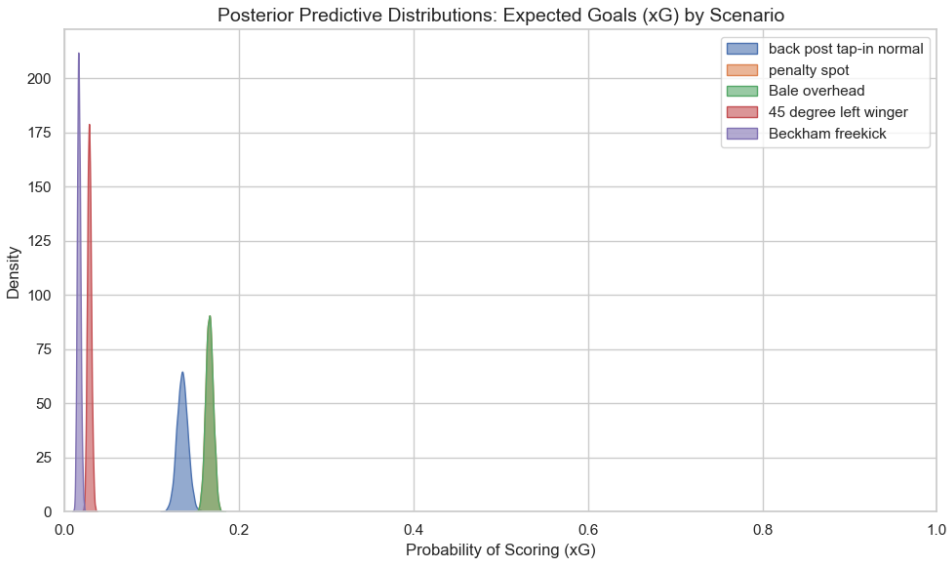

Figure 8: Model 1 Predictive Density

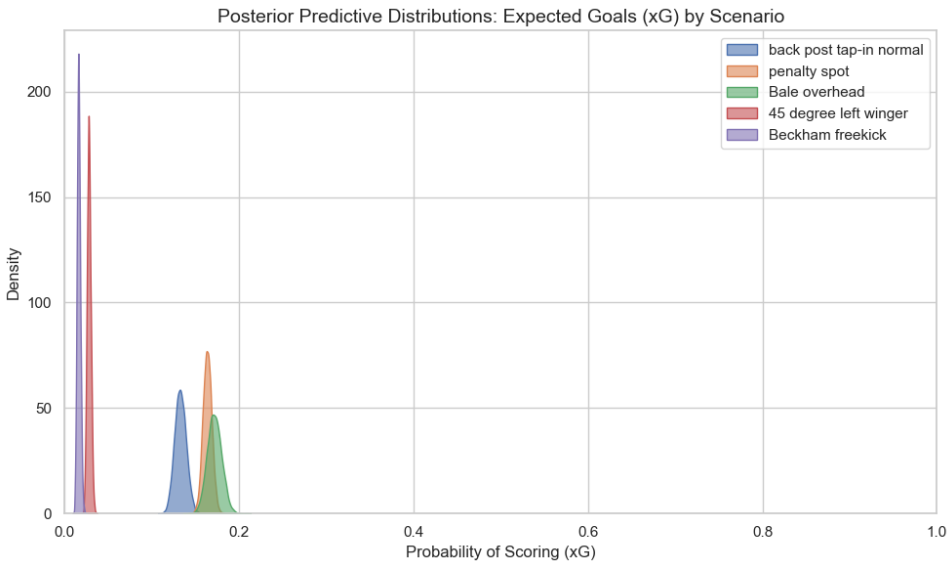

Figure 9: Model 2 Predictive Density

References

[1] StatsBomb. *StatsBomb Open Data*. GitHub repository, 2024. https://github.com/statsbomb/statsbombpy

[2] Salvatier, J., Wiecki, T. V., and Fonnesbeck, C. *Probabilistic programming in Python using PyMC3*. PeerJ Computer Science, 2(e55), 2016. https://www.pymc.io/welcome.html

[3] Hoffman, M. D. and Gelman, A. *The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo*. Journal of Machine Learning Research, 15(1):1593–1623, 2014.

[4] UEFA Champions League Final, Real Madrid vs. Liverpool. Gareth Bale overhead kick goal, Kyiv, 2018. https://youtu.be/VfhFW18uhBE

[5] Premier League, Tottenham Hotspur vs. Brighton & Hove Albion, Heung-Min Son goal, London, 8 April 2023. https://youtu.be/f18OxcZ6mI4

[6] FIFA World Cup Qualifier, England vs. Greece. David Beckham long-range free kick, Old Trafford, 2001. https://youtu.be/tOGESlaVNdE

[7] Dixon, M. J. and Coles, S. G. *Modelling Association Football Scores and Inefficiencies in the Football Betting Market*. Journal of the Royal Statistical Society: Series C (Applied Statistics), 46(2):265–280, 1997.

[8] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. *Bayesian Data Analysis*, 3rd ed. Chapman and Hall/CRC, 2013.